Universal gravitation

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Predict how gravitational force changes with mass and separation.

SUCCESS CRITERIA - I can...

1. Use center-to-center separation.
2. Apply direct mass and inverse-square distance comparisons.
3. Identify equal forces on the two interacting bodies.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “gravity” and “separation” to explain your model or result. Sentence starters are optional.

gravity: an attractive interaction between masses

separation: distance between the specified objects or centers

inverse square: proportional to one divided by distance squared

01 Bring back an earlier idea

A 3 kg object moves at 5 m/s in an ideal circle under a force perpendicular to velocity.
Explain which parts of velocity change and whether the force does work.

Word help: ideal = using stated simplifying conditions

Sentence starter: Under the assumption ..., the evidence ... supports ... because ... .

Marzano check - circle: ① ② ③

02 Explain a worked example

Reference: $F = Gm_1m_2/r^2$.
r is center-to-center separation. At fixed masses, doubling r makes the attractive force one quarter as large.

Worked reasoning: If separation doubles while both masses stay fixed, gravitational force becomes $1/2^2=1/4$ as large. A 20 N interaction would become 5 N.

Your turn: Sketch the two separations using centers.
Draw equal and opposite force arrows for each pair.

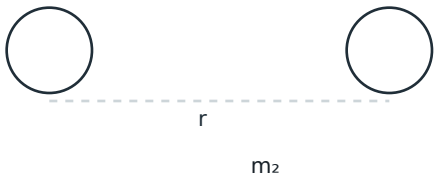
Sentence starter: The ... in my model represents ... . Its label shows ... .

Marzano check - circle: ① ② ③

03 Represent and interpret

Two equal spheres are shown.
Mark the distance r used by the gravity equation and draw a force on each sphere.

Sentence starter: The ... in my model represents ... . Its label shows ... .



Each circle represents one of the labeled objects or charges; r is their center-to-center separation.

Marzano check - circle: ① ② ③

Read down the left column, then down the right. Use the open space to draw, calculate and explain.

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Continue down each column.

Sentence starters are optional. You may plan in any language.

04 Solve and justify

Starting from force F_0, triple one mass, double the other, and triple r.
Find F/F_0 and justify every factor.

Word help: justify = support your answer with a reason or evidence

Sentence starter: Under the assumption ..., the evidence ... supports ... because ... .

Marzano check - circle: 1 2 3

05 Analyze the reasoning

Does a large planet exert more force on a small moon than the moon exerts on the planet?
Compare forces and accelerations.

Sentence starter: Both ... . They differ in ... because ... .

Marzano check - circle: 1 2 3

06 Choose a method

A spacecraft of mass 4 kg pushes an asteroid of mass 400 kg with 24 N.
Compare interaction forces and the accelerations due to that interaction.

Word help: interaction = two objects affecting one another

Sentence starter: Both ... . They differ in ... because ... .

Marzano check - circle: 1 2 3

07 Exit check

What separation makes force 1/9 of its original value? What if both masses also triple?

Sentence starter: I notice ... . This supports ... because ... .

Marzano check - circle: 1 2 3

08 STRETCH: Apply it another way

Both masses triple. What separation multiplier keeps the force unchanged? What multiplier makes it half the original force?

Sentence starter: I notice ... . This supports ... because ... .

Marzano check - circle: 1 2 3